Evaluating Sequence-Generating Architectures

Sequence-generating models, such as those used in machine translation, summarization, and dialogue systems, produce outputs that require specialized evaluation metrics beyond simple accuracy. This document explores the challenges of assessing generated sequences, contrasts intrinsic and extrinsic evaluation methods, and provides detailed formulations of key metrics—Perplexity, BLEU, ROUGE, METEOR, and CHRF—along with examples and comparisons. We aim to equip researchers with a comprehensive toolkit for evaluating seq2seq performance across diverse tasks.

1 Introduction

Sequence-to-sequence (seq2seq) models, including Transformers, RNNs, LSTMs, and GRUs, generate sequences like translations, summaries, or chatbot responses. Unlike classification tasks with a single correct label, sequence generation permits multiple valid outputs, complicating evaluation. This document addresses:

1. Challenges in evaluating generated sequences,
2. Intrinsic (automated) vs. extrinsic (human) evaluation approaches,
3. Mathematical formulations of widely used metrics,
4. Practical examples of metric calculations,
5. Comparative analysis of metrics and their applications.

2 Challenges in Evaluating Sequence Generation

Evaluating seq2seq models is inherently complex due to the following issues:

2.1 Multiple Correct Outputs

A single input can have several valid outputs. For example, in machine translation: - **Input**: "Il fait beau aujourd'hui." - **Reference 1**: "The weather is nice today." - **Reference 2**: "It's a beautiful day today." - **Generated**: "Today is lovely weather." The generated output is semantically correct but differs lexically, challenging word-overlap metrics.

2.2 Fluency vs. Accuracy Tradeoff

Generated sequences must balance grammatical fluency and semantic accuracy. Example: - **Reference**: "The cat slept on the sofa." - **Generated**: "A feline rested on the couch." While fluent and accurate, lexical differences may lower scores in precision-based metrics.

2.3 Long-Form Generation Issues

Long sequences often exhibit: - **Repetition**: "The cat cat cat sat." - **Incoherence**: "The cat flew to the moon sofa." - **Factual Errors**: "The cat slept on the sun." Metrics must detect these flaws beyond simple overlap.

3 Intrinsic vs. Extrinsic Evaluation

3.1 Intrinsic Evaluation (Automated Metrics)

Automated metrics compute scores based on mathematical comparisons to reference sequences. - **Pros**: Fast, scalable, reproducible. - **Cons**: May not capture fluency, coherence, or human perception. - **Examples**: BLEU, ROUGE, Perplexity.

3.2 Extrinsic Evaluation (Human Judgment)

Human evaluators score outputs on fluency, coherence, and relevance (e.g., 1–5 scale). - **Pros**: Captures subjective quality; gold standard for creative tasks. - **Cons**: Expensive, time-consuming, subjective. - **Example**: Rating a chatbot response for naturalness.

4 Mathematical Formulation of Key Metrics

4.1 Perplexity (PPL)

Perplexity quantifies how well a model predicts a sequence, reflecting the uncertainty in token prediction:

$$\mathrm{PPL} = \exp\left(-\frac{1}{N}\sum_{t=1}^{N}\log P(y_t \mid y_{<t}, X)\right),$$

- $P(y_t \mid y_{<t}, X)$: Probability of token y_t given prior tokens and input X, - N: Total tokens in the test set. - **Range**: $[1, \infty)$ (lower is better). - **Interpretation**: Lower PPL indicates better fit to the data. - **Limitation**: Requires ground truth; doesn't assess generated output quality directly.

Example: For a sequence "I am happy" with probabilities $P(\mathrm{I}) = 0.9$, $P(\mathrm{am} \mid \mathrm{I}) = 0.8$, $P(\mathrm{happy} \mid \mathrm{I\ am}) = 0.7$:

$$\mathrm{PPL} = \exp\left(-\frac{1}{3}(\log 0.9 + \log 0.8 + \log 0.7)\right) \approx 1.36.$$

4.2 BLEU (Bilingual Evaluation Understudy)

BLEU measures n-gram precision with a brevity penalty:

$$\mathrm{BLEU} = \mathrm{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right),$$

- p_n: Precision for n-grams:

$$p_n = \frac{\sum_{\text{n-gram}\in\hat{Y}} \min(\mathrm{Count}_{\hat{Y}}(\text{n-gram}), \max_{\mathrm{ref}} \mathrm{Count}_{\mathrm{ref}}(\text{n-gram}))}{\sum_{\text{n-gram}\in\hat{Y}} \mathrm{Count}_{\hat{Y}}(\text{n-gram})},$$

- BP: Brevity penalty:

$$\mathrm{BP} = \begin{cases} 1, & \text{if } c > r, \\ e^{1-r/c}, & \text{if } c \leq r, \end{cases}$$

where $c = |\hat{Y}|$, $r = |\text{ref}|$. - w_n: Weights (e.g., 1/4 for BLEU-4). - **Range**: 0–1 (higher is better).

Example: - **Reference**: "The cat sat on the mat." - **Generated**: "The cat is sitting on the mat." - Unigrams: $p_1 = 6/7 \approx 0.857$ (all but "is" match), - Bigrams: $p_2 = 3/6 = 0.5$ ("The cat", "on the", "the mat"), - Trigrams: $p_3 = 1/5 = 0.2$ ("on the mat"), - BP = 1 ($7 > 6$), - BLEU-2: $\exp\left(\frac{1}{2}(\log 0.857 + \log 0.5)\right) \approx 0.654$.

4.3 ROUGE (Recall-Oriented Understudy for Gisting Evaluation)

ROUGE-N measures recall of n-grams:

$$\text{ROUGE-N} = \frac{\sum_{\text{ref} \in R} \sum_{\text{n-gram} \in \text{ref}} \text{Count}_{\text{match}}(\text{n-gram})}{\sum_{\text{ref} \in R} \sum_{\text{n-gram} \in \text{ref}} \text{Count}(\text{n-gram})},$$

- **Range**: 0–1 (higher is better). - **Variants**: ROUGE-L (longest common subsequence), ROUGE-S (skip bigrams).

Example: - **Reference**: "The quick brown fox jumps." - **Generated**: "Quick fox leaps fast." - ROUGE-1: $3/5 = 0.6$ ("quick", "fox", "jumps/leaps" match), - ROUGE-2: $1/4 = 0.25$ ("quick fox").

4.4 METEOR

METEOR uses precision, recall, and a word-order penalty:

$$\text{METEOR} = F_{\text{mean}} \cdot (1 - \text{Penalty}),$$

- $F_{\text{mean}} = \frac{10PR}{R+9P}$, - $P = \frac{\text{\# matched unigrams}}{\text{\# unigrams in generated}}$, - $R = \frac{\text{\# matched unigrams}}{\text{\# unigrams in reference}}$, - Penalty: $0.5 \cdot \left(\frac{\text{\# chunks}}{\text{\# matches}}\right)^3$. - **Range**: 0–1 (higher is better).

Example: - **Reference**: "The cat sleeps." - **Generated**: "Cat is sleeping." - Matches: "cat", "sleeps/sleeping" (synonym), $P = 2/3$, $R = 2/3$, - $F_{\text{mean}} = \frac{10 \cdot 2/3 \cdot 2/3}{2/3 + 9 \cdot 2/3} \approx 0.667$, - Penalty ≈ 0 (one chunk), METEOR ≈ 0.667.

4.5 CHRF (Character-Level F-score)

$$\text{CHRF} = \frac{2 \cdot P_{\text{char}} \cdot R_{\text{char}}}{P_{\text{char}} + R_{\text{char}}},$$

- $P_{\text{char}}, R_{\text{char}}$: Precision and recall of character n-grams. - **Range**: 0–1 (higher is better).

Example: - **Reference**: "cat", - **Generated**: "cats", - Character trigrams: Ref: "cat"; Gen: "cat", "ats", - $P = 1/2 = 0.5$, $R = 1/1 = 1$, CHRF $= \frac{2 \cdot 0.5 \cdot 1}{0.5+1} \approx 0.667$.

5 Examples of Metric Calculations

5.1 High Similarity Case

- **Reference**: "The sun shines brightly." - **Generated**: "The sun glows bright." - **BLEU-2**: $p_1 = 4/5 = 0.8$, $p_2 = 2/4 = 0.5$, BP $= e^{1-5/5} = 1$, BLEU ≈ 0.632. - **ROUGE-1**: $4/5 = 0.8$. - **METEOR**: $P = 4/5$, $R = 4/5$, $F_{\text{mean}} = 0.8$, Penalty ≈ 0, METEOR ≈ 0.8.

5.2 Low Similarity Case

- **Reference**: "The sun shines brightly." - **Generated**: "Rain falls heavily now." - **BLEU**: $p_1 = 0/5 = 0$, BLEU $= 0$. - **ROUGE-1**: $0/5 = 0$. - **METEOR**: No matches, METEOR ≈ 0.

6 Comparison of Metrics

Metric	Best For	Synonyms	Paraphrasing	Fluency
Perplexity	Language Modeling	No	No	No
BLEU	Translation	No	No	No
ROUGE	Summarization	No	No	No
METEOR	Translation, Chatbots	Yes	Yes	Partial
CHRF	Rich Morphology	Partial	Partial	No

Table 1: Comparison of evaluation metrics for sequence generation.

7 Conclusion

- No single metric fully captures seq2seq performance; combining BLEU, METEOR, and ROUGE offers a balanced view. - Human evaluation remains essential for fluency and coherence, especially in creative tasks. - Task-specific selection is key: BLEU for translation, ROUGE for summarization, METEOR for semantic tasks.